THE EXACT SEQUENCE OF MODULES OF DIFFERENTIALS FOR A COMPOSITION OF MORPHISMS OF RINGED SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The functoriality of the module of differentials for a morphism of ringed spaces, together with the description of the pullback of a module on stalks. Both are invoked inside the counted proof (source lines 5341-5342, 5354-5361) as pointers to a lemma of this chapter and a result of another chapter. Identifies the maps of the sequence with the maps appearing in the algebraic exact sequence.
- The compatibility of the formation of differentials with taking stalks. Invoked inside the counted proof at source line 5347 as a pointer to a lemma of this chapter. Turns the sequence of sheaves into a sequence of modules over the local rings of the stalks.

Source reference. Original lemma lemma-functoriality-differentials-ringed-spaces: sources/stacks-modules/pinned-original-source.tex, source lines 5323–5323. Navigation pointer only: the counted text refers to the Stacks item lemma-functoriality-differentials-ringed-spaces, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-stalk-module-differentials: sources/stacks-modules/pinned-original-source.tex, source lines 5197–5197. Navigation pointer only: the counted text refers to the Stacks item lemma-stalk-module-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference algebra-lemma-exact-sequence-differentials: sources/stacks-modules/pinned-original-source.tex, source lines 5355–5355. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-exact-sequence-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-lemma-stalk-pullback-modules: sources/stacks-modules/pinned-original-source.tex, source lines 1418–1418. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-stalk-pullback-modules, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$, $g : Y \to S$ be morphisms of ringed spaces Then there is a canonical exact sequence*

$$f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$$

where the maps come from applications of Lemma lemma-functoriality-differentials-ringed-spaces.

Proof. By taking induced maps in stalks at $x \in X$ and using Lemma lemma-stalk-module-differentials we obtain a sequence

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,f(x)}} \Omega_{\mathcal{O}_{Y,f(x)}/\mathcal{O}_{S,g(f(x))}} \to \Omega_{\mathcal{O}_{X,x}/\mathcal{O}_{S,g(f(x))}} \to \Omega_{\mathcal{O}_{X,x}/\mathcal{O}_{Y,f(x)}} \to 0$$

It suffices to see that the maps of the sequence are the same as the ones in Algebra, Lemma algebra-lemma-exact-sequence-differentials. This is because the characterization of the maps in Lemma lemma-functoriality-differentials-ringed-spaces and because via the isomorphism of Sheaves, Lemma sheaves-lemma-stalk-pullback-modules we have $(f^*s)_x = s_x \otimes 1$, for any local section s of a sheaf of $\mathcal{O}_Y$-modules. □